19) Experiment to understand array function in Data science.

Program:

```
import numpy as np
array = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90])
print(array)
print(np.sqrt(array))
print(array.ndim)
new_array = array.reshape(3, 3)
print(new_array)
print(new_array.ndim)
print(new_array.ravel())
newm = new_array.reshape(3, 3)
print(newm[2, 1:3])
print(newm[1:2, 1:3])
print(new_array[0:3, 0:1])
print(new_array[1:3])
```

Output:

```
[10 20 30 40 50 60 70 80 90]
[3.16227766 4.47213595 5.47722558 6.32455532 7.07106781 7.74596669
 8.36660027 8.94427191 9.48683298]
1
[[10 20 30]
 [40 50 60]
 [70 80 90]]
2
[10 20 30 40 50 60 70 80 90]
[80 90]
[[50 60]]
[[10]
 [40]
 [70]]
[[40 50 60]
 [70 80 90]]
```

Result:

Thus, NumPy array functions such as square root, dimensions, reshaping, flattening, and slicing were performed successfully.